KungFlow

Executive Summary

Protect your focus

Ido Mark · Rotem Saraf · Eden Elkayam 2026

1. Overview

KungFlow is an adaptive focus-protection system for employees whose work is primarily computer-based and frequently interrupted by digital notifications. It runs as a Windows desktop product that watches behavioral work patterns such as typing rhythm, mouse movement, and application or window switching. These signals are evaluated relative to the user's personal behavioral baseline.

The core product logic is Sense, Understand, Protect. KungFlow senses behavioral rhythm, understands whether the current work pattern is elevated relative to the user's baseline, and can activate a notification firewall when the system detects elevated cognitive load. The goal is to protect focus at sensitive moments without requiring the user to remember to enable Do Not Disturb manually.

KungFlow addresses two connected audiences. Employees are the direct users and experience fragmented focus, stress, and difficulty maintaining deep work. Their organizations are the potential business customers, because repeated focus disruption reduces productive work time and work efficiency. The current implementation is a working Windows-based client-server system using C# and .NET for the desktop agent, Node.js and Express for the backend, Docker, Google Cloud Run, and Google Cloud SQL.

2. The Problem, Target Audience & Validation

THE PROBLEM

Digital work environments expose employees to frequent notifications, messages, alerts, application switching, and context switching. The problem is not simply that notifications are annoying. The deeper issue is that interruptions repeatedly break concentration, and an interruption that arrives when a worker is already cognitively loaded can be especially disruptive.

Evidence	Meaning for KungFlow
56 interruptions per workday	Frequent pings, messages, and alerts create repeated pressure on attention.
Approximately one interruption every 9 minutes	Focus is disrupted at a rhythm that makes sustained concentration difficult.
23 minutes to regain deep focus	Even short interruptions can create long recovery costs for focused work.
26% higher stress / lower job satisfaction	Digital interruptions are linked with stress, frustration, and time pressure.

TARGET AUDIENCE

KungFlow serves two closely connected audiences: employees are the direct users of the product, while the organizations employing them are the potential business customers. For employees, the impact includes fragmented focus, higher stress, and difficulty maintaining sustained concentration. For companies, the impact is qualitative but important: repeated interruptions reduce productive time and work efficiency.

RESEARCH-BASED VALIDATION

The project uses three distinct layers of validation. First, academic research documented in the project README supports the problem definition: digital interruptions can disrupt focus, context switching is associated with cognitive load, and frequent digital interruptions can be associated with stress and reduced uninterrupted focus.

Second, the literature review supports the relevance of observable computer-interaction behaviors. Mouse movement, typing and keystroke patterns, and application or context switching can provide indicators related to cognitive load. These findings provided the foundation for selecting KungFlow's behavioral signals, but they do not validate the exact scoring algorithm.

Third, the team conducted an initial controlled pilot with n = 8 participants, a learning-effect control group, and an error-correction task comparing work with interruptions against work with reduced or removed interruptions. The results showed +165% better task scores once interruptions were removed, 49% fewer missed errors in the same 10-minute window, and a +25% net gain after controlling for natural learning effects. Because the sample is small, this is initial validation of the interruption-reduction hypothesis, not proof of organizational productivity gains or validation of the cognitive-load detection algorithm itself.

Full academic references supporting the problem definition and behavioral-signal selection are available in the project README on GitHub.

3. Product & Value Proposition

KungFlow runs in the background on Windows and collects behavioral metadata rather than user content. The desktop agent measures work-pattern signals in short windows and sends metadata samples to the server. The server stores the metrics, updates the user's cognitive state, and returns a protection decision to the desktop agent.

Signal group	Examples used in the project	Purpose
Mouse behavior	Speed, acceleration, micro-pauses	Indicates changes in interaction rhythm during real work.
Typing behavior	Typing rhythm, pace changes, delete / backspace activity	Captures friction without storing typed content.
Application / window switching	Application switching, window switching, visible / open-window patterns	Represents context switching and attention fragmentation.

The product moves through three visible states. Open indicates low cognitive load and allows notifications to pass. Calibrating means KungFlow is learning the user's normal behavioral baseline before full automation. Firewall Active means elevated load was detected and focus protection has been activated.

Protection follows the user's selected configuration. The current product supports manual and automatic control, global notification protection, and selective protection for available application notification settings. The value to employees is less disruption at sensitive moments, less need to consciously manage Do Not Disturb, and automatic focus protection while they work normally. The existing product message captures this simply: No buttons. No willpower. No new habits.

For companies, KungFlow offers a way to support productive work without relying on intrusive surveillance. It aims to improve work efficiency by protecting focus, while avoiding content capture such as typed text, visited websites, opened documents, or screenshots.

IMPLEMENTED SYSTEM AT A GLANCE

Layer	Technology	Role
Windows Desktop Agent	C# · .NET	Collects behavioral metadata and applies the returned firewall decision.
KungFlow Server	Node.js · Express · Docker · Google Cloud Run	Receives samples, scores current load against the user's baseline, and returns protection decisions.
SQL Database	Google Cloud SQL	Stores users, sessions, metrics, baseline state, current state, and firewall events.

4. Innovation & Key Advantages

KungFlow's innovation is not simply muting notifications. Operating systems already provide Do Not Disturb. The product connects behavioral sensing, personalized interpretation, and automatic intervention into one continuous workflow.

- Automatic real-time intervention: KungFlow is designed to act when elevated load is detected, rather than only reporting activity after the fact.
- Personal / adaptive baseline: the system evaluates each user relative to their own work rhythm instead of assuming one universal behavior level fits everyone.
- Research-backed behavioral sensing: the selected signals are grounded in prior research linking aspects of computer interaction behavior with cognitive load.
- Privacy by design: KungFlow uses behavioral metadata and does not need typed text, visited websites, opened documents, or screenshots.

This combination keeps the product focused on employee support rather than surveillance. The system is designed to help the user work normally while reducing interruptions during moments of elevated load.

5. Competitors & Alternative Solutions

Alternative	Main orientation	KungFlow differentiation
RescueTime	Reports activity and productivity after the fact.	Aims to detect and protect in real time.
Calm for Business	Supports wellbeing outside the immediate workflow.	Operates inside the work context where interruptions occur.
Microsoft Viva Insights	Provides productivity and organizational insights.	Connects overload sensing to individual protection.
Built-in Do Not Disturb	Can reduce interruptions when the user activates it.	Aims to decide when protection is needed automatically.

KungFlow's position is the missing layer between passive analytics and manual notification controls: it aims to detect when protection is needed and apply it automatically in real time.

6. Target Market & Market Potential

The broader market consists of knowledge workers whose work is primarily digital. The current commercial focus is Israeli technology companies, which employ many of the computer-based employees who represent the direct users of the product.

Market scope	Market figure	**Interpretation**
TAM	1.2B knowledge workers worldwide	Broad population of digital workers exposed to interruption-heavy environments.
SAM	400K technology workers in Israel	Initial reachable segment for the team and local technology market.
SOM Y1	2,500 users across 10 mid-market companies	Focused first-year adoption target from the current presentation.

In this framing, users are employees and business customers are the organizations that employ them.

7. Go-to-Market Strategy

The go-to-market plan follows the simplified current presentation story. It begins with design partners from the team's existing network, moves into founder-led sales, and then scales outbound once early validation and measurable case-study material exist.

Phase	Action	Purpose
Phase 1	Design partners	Validate real-world use, collect feedback, and learn from actual team usage.
Phase 2	Founder-led sales	Direct outreach to VP R&D and VP People at Israeli technology companies.
Phase 3	Scale outbound	Expand the outreach process more systematically after early validation.

Technology and product next steps include scaling the existing implementation, continuing v2 development, and conducting larger validation with design partners.

8. Team & Responsibilities

The current KungFlow project team consists of Ido Mark, Rotem Saraf, and Eden Elkayam.

Team member	Responsibility
Ido Mark	Team Lead · Problem validation · Backend/server development · Windows desktop development · Cognitive-score formula design
Rotem Saraf	Problem validation · Backend/server development · Desktop UI · Database design & implementation · Cloud deployment
Eden Elkayam	Problem validation · Database design & implementation